

M. PHARM (PHARMACOLOGY)
(SEM I) THEORY EXAMINATION 2022-23
MODERN PHARMACEUTICAL ANALYTICAL TECHNIQUES

Time: 3 Hours**Total Marks: 75****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A**

1. Attempt all questions in brief. 10 x 2 = 20

- (a) Name the factors affecting vibrational frequencies in IR spectroscopy.
- (b) Compare fluorescence and phosphorescence.
- (c) State the importance of quantum numbers in NMR.
- (d) Outline the role of internal standard in NMR. Give example.
- (e) What is the role of analyzers in mass spectrometry?
- (f) Write the advantages of HPTLC over TLC.
- (g) Discuss the principle of affinity chromatography.
- (h) Name the factors affecting separation in Electrophoresis.
- (i) Detail the applications of potentiometry.
- (j) Give the principle of TGA.

SECTION B

2. Attempt any two parts of the following: 2 x 10 = 20

- (a) Explain the principle, instrumentation, interferences and applications of Atomic absorption spectrometry.
- (b) Discuss the principle, instrumentation, and applications of NMR. Add a note on nuclear magnetic double resonance.
- (c) Outline Bragg's Law in detail. Explain the production of X-Rays and Rotating crystal technique.

SECTION C

3. Attempt any five parts of the following: 5 x 7 = 35

- (a) Discuss Lambert-Beer's Law.
- (b) What is chemical shift? Discuss various factors affecting chemical shift.
- (c) Explain Quadrupole and Time of Flight Mass Analyzers.
- (d) Illustrate instrumentation of Gas chromatography with special emphasis on detectors used in the technique.
- (e) Elaborate principle, instrumentation and applications of paper electrophoresis.
- (f) Discuss principle and types of ELISA.
- (g) Illustrate sample handling techniques in IR and instrumentation of FT-IR.